

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404746
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lorenson; Troy et al.

Flow-through pulsing assembly for use in downhole operations

Abstract

A flow-through assembly for use in a downhole drilling string includes a Moineau-type motor, means for selectively activating the motor such as a ball catch component that selectively causes drilling fluid to enter into or bypass the motor, and a rotating variable choke assembly that is driven by a rotor of the motor. The choke assembly varies the flow rate of drilling fluid as rotation causes ports of the choke assembly to enter into and out of alignment with each other. In one embodiment, the choke assembly comprises a faceted rotary component including bypass ports on the facets of the component. In another embodiment, the choke assembly comprises a tapered rotary component that rotates in a complementarily tapered stationary component.

Inventors: Lorenson; Troy (Edmonton, CA), Leroux; Kevin (Beaumont, CA), Parenteau; Dwayne (Edmonton, CA), Kinsella; Douglas (Sturgeon County, CA)

Applicant: IMPULSE DOWNHOLE SOLUTIONS LTD. (Edmonton, CA)

Family ID: 60901431

Assignee: IMPULSE DOWNHOLE SOLUTIONS LTD. (Edmonton, CA)

Appl. No.: 18/460520

Filed: September 01, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230417126 A1	Dec. 28, 2023

Related U.S. Application Data

continuation parent-doc US 17197896 20210310 US 11788382 child-doc US 18460520
continuation parent-doc US 16241029 20190107 US 10968721 20210406 child-doc US 17197896

Publication Classification

Int. Cl.: **E21B7/24** (20060101); **E21B4/02** (20060101); **E21B34/06** (20060101); **E21B34/14** (20060101); **F03C2/08** (20060101); **F04C2/107** (20060101); **F04C13/00** (20060101); **F04C14/24** (20060101)

U.S. Cl.:

CPC **E21B34/142** (20200501); **E21B4/02** (20130101); **E21B34/06** (20130101); **F03C2/08** (20130101); **F04C2/107** (20130101); **F04C13/008** (20130101); **F04C14/24** (20130101);

Field of Classification Search

CPC: E21B (47/185); E21B (28/00); E21B (43/003); E21B (4/02); E21B (7/24); F16K (3/085)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
860684	12/1906	Montgomery	N/A	N/A
1132063	12/1914	Bardeen	N/A	N/A
2250912	12/1940	Hudson	N/A	N/A
2329912	12/1942	Kent	N/A	N/A
2481059	12/1948	Africano	N/A	N/A
2569026	12/1950	Springer	N/A	N/A
2738956	12/1955	Bielstein	N/A	N/A
2743083	12/1955	Zublin	N/A	N/A
2746721	12/1955	Moore	N/A	N/A
2780438	12/1956	Bielstein	N/A	N/A
2836395	12/1957	Bielstein	N/A	N/A
2896916	12/1958	Clavier	N/A	N/A
2942851	12/1959	Beck	N/A	N/A
3065416	12/1961	Jeter	N/A	N/A
3096833	12/1962	Bodine	N/A	N/A
3101796	12/1962	Stall et al.	N/A	N/A
3216514	12/1964	Nelson	N/A	N/A
3243001	12/1965	Vincent	N/A	N/A
3252689	12/1965	Blomgren	N/A	N/A
3270822	12/1965	Cleary	N/A	N/A
3292898	12/1965	Willman	N/A	N/A
3547006	12/1969	Rudman	N/A	N/A
3557875	12/1970	Solum et al.	N/A	N/A
3640351	12/1971	Coyne	N/A	N/A
3739331	12/1972	Godbey et al.	N/A	N/A
3768576	12/1972	Martini	N/A	N/A
3894818	12/1974	Tschirky	N/A	N/A

3899033	12/1974	Van Huisen	N/A	N/A
3933209	12/1975	Sweeney	N/A	N/A
3941196	12/1975	Curington	N/A	N/A
4033429	12/1976	Farr	N/A	N/A
4058163	12/1976	Yandell	N/A	N/A
4080115	12/1977	Sims	N/A	N/A
4187061	12/1979	Jurgens	418/153	F04C 15/0042
4291723	12/1980	Beimgraben	N/A	N/A
4296822	12/1980	Ormsby	N/A	N/A
4360040	12/1981	Cove	N/A	N/A
4384625	12/1982	Roper	N/A	N/A
4499956	12/1984	Campbell	N/A	N/A
4574894	12/1985	Jadwin	137/71	E21B 34/142
4705117	12/1986	Warren	N/A	N/A
4789032	12/1987	Rehm	N/A	N/A
4817739	12/1988	Jeter	N/A	N/A
4819745	12/1988	Walter	N/A	N/A
4830122	12/1988	Walter	N/A	N/A
4890682	12/1989	Worrall	N/A	N/A
4936397	12/1989	McDonald et al.	N/A	N/A
4953595	12/1989	Kotlyar	N/A	N/A
4979577	12/1989	Walter	N/A	N/A
5009272	12/1990	Walter	N/A	N/A
5048622	12/1990	Ide	N/A	N/A
5139400	12/1991	Ide	N/A	N/A
5176164	12/1992	Boyle	N/A	N/A
5186614	12/1992	Abousabha	N/A	N/A
5190114	12/1992	Walter	N/A	N/A
5404945	12/1994	Head	138/42	E21B 21/10
5513713	12/1995	Groves	N/A	N/A
5535835	12/1995	Walker	N/A	N/A
5662180	12/1996	Coffman	N/A	N/A
5673751	12/1996	Head	N/A	N/A
5692563	12/1996	Krueger	N/A	N/A
6199586	12/2000	Pawelzik	N/A	N/A
6279670	12/2000	Eddison	N/A	N/A
6289911	12/2000	Majkovic	N/A	N/A
6431294	12/2001	Eddison	N/A	N/A
6439318	12/2001	Eddison	N/A	N/A
6469637	12/2001	Seyler et al.	N/A	N/A
6508317	12/2002	Eddison	N/A	N/A
6571870	12/2002	Zheng	N/A	N/A
6866104	12/2004	Stoesz	N/A	N/A
6994175	12/2005	Egerstrom	N/A	N/A
7073610	12/2005	Susman	N/A	N/A
7139219	12/2005	Kolle	N/A	N/A
7163058	12/2006	Bakke	N/A	N/A
7219752	12/2006	Wassell	N/A	N/A
7424922	12/2007	Hall	N/A	N/A

7549485	12/2008	Radford et al.	N/A	N/A
7575051	12/2008	Stoesz et al.	N/A	N/A
7708088	12/2009	Allahar et al.	N/A	N/A
7921937	12/2010	Brackin	N/A	N/A
8162078	12/2011	Anderson	N/A	N/A
8167051	12/2011	Eddison	N/A	N/A
8181719	12/2011	Bunney	N/A	N/A
8201641	12/2011	Allahar	N/A	N/A
8322463	12/2011	Walter	N/A	N/A
8535028	12/2012	Groves	N/A	N/A
8636073	12/2013	McNeilly	N/A	N/A
8869916	12/2013	Clausen et al.	N/A	N/A
9004194	12/2014	Eddison	N/A	N/A
9091123	12/2014	Gust	N/A	N/A
9181767	12/2014	Schultz	N/A	N/A
9212522	12/2014	Schultz	N/A	N/A
9267539	12/2015	Gynz-Rekowski	N/A	N/A
9273529	12/2015	Eddison et al.	N/A	N/A
9359866	12/2015	Mackenzie	N/A	N/A
9371692	12/2015	Eddison	N/A	N/A
9382760	12/2015	Le	N/A	N/A
9382950	12/2015	Pheasey	N/A	N/A
9476263	12/2015	Clausen et al.	N/A	N/A
9494006	12/2015	Mohon et al.	N/A	N/A
9523251	12/2015	Honekamp	N/A	N/A
9540877	12/2016	Lanning et al.	N/A	N/A
9593537	12/2016	Gust	N/A	N/A
9593547	12/2016	Malcolm	N/A	N/A
9598923	12/2016	Gilleylen et al.	N/A	N/A
9624767	12/2016	Sitka	N/A	N/A
9637976	12/2016	Lorenson	N/A	N/A
9732573	12/2016	Adam	N/A	N/A
9752411	12/2016	Adam	N/A	N/A
9765584	12/2016	Lorenson et al.	N/A	N/A
9840873	12/2016	Schultz	N/A	N/A
9879495	12/2017	Baudoin	N/A	N/A
9932774	12/2017	Gillis	N/A	N/A
9957765	12/2017	Schultz	N/A	N/A
10316588	12/2018	Robertson et al.	N/A	N/A
10358872	12/2018	Rossing et al.	N/A	N/A
10590709	12/2019	Sicilian	N/A	N/A
10633920	12/2019	Lorenson et al.	N/A	N/A
10648265	12/2019	Kinsella et al.	N/A	N/A
10724318	12/2019	Schultz	N/A	N/A
10927601	12/2020	Lorenson	N/A	N/A
10989004	12/2020	Russell et al.	N/A	N/A
11268337	12/2021	Kinsella et al.	N/A	N/A
2002/0157871	12/2001	Tulloch	N/A	N/A
2004/0045716	12/2003	Bakke	N/A	N/A
2005/0126828	12/2004	Pinol	N/A	N/A

2005/0211471	12/2004	Zupanick	N/A	N/A
2006/0108152	12/2005	Lee	N/A	N/A
2006/0237187	12/2005	Stoesz et al.	N/A	N/A
2006/0243493	12/2005	El-Rayes	N/A	N/A
2006/0272821	12/2005	Webb	N/A	N/A
2007/0221409	12/2006	Hall	N/A	N/A
2007/0221412	12/2006	Hall	N/A	N/A
2008/0000697	12/2007	Rytlewski	N/A	N/A
2008/0029268	12/2007	Macfarlane	N/A	N/A
2008/0135295	12/2007	Hall	N/A	N/A
2008/0271923	12/2007	Kusko	N/A	N/A
2009/0095528	12/2008	Hay	N/A	N/A
2009/0101328	12/2008	Leslie et al.	N/A	N/A
2009/0107723	12/2008	Kusko	N/A	N/A
2009/0183919	12/2008	Hall	N/A	N/A
2009/0266612	12/2008	Allahar et al.	N/A	N/A
2010/0018201	12/2009	Mochizuki	N/A	N/A
2010/0065334	12/2009	Hall	N/A	N/A
2010/0212900	12/2009	Eddison	N/A	N/A
2010/0212901	12/2009	Buytaert	N/A	N/A
2010/0224412	12/2009	Allahar	N/A	N/A
2010/0247359	12/2009	Hauri	N/A	N/A
2010/0276198	12/2009	Light	N/A	N/A
2010/0326733	12/2009	Anderson	N/A	N/A
2011/0000716	12/2010	Comeau	N/A	N/A
2011/0073374	12/2010	Bunney et al.	N/A	N/A
2011/0240315	12/2010	McNeilly	N/A	N/A
2012/0048619	12/2011	Seutter	N/A	N/A
2012/0193145	12/2011	Anderson	N/A	N/A
2012/0279724	12/2011	Eddison et al.	N/A	N/A
2013/0048386	12/2012	Le et al.	N/A	N/A
2013/0118812	12/2012	Clausen et al.	N/A	N/A
2013/0168099	12/2012	Themig	N/A	N/A
2013/0251512	12/2012	Lombard	N/A	N/A
2013/0277116	12/2012	Knoll	N/A	N/A
2014/0041943	12/2013	Lanning	N/A	N/A
2014/0048283	12/2013	Mohon et al.	N/A	N/A
2014/0060830	12/2013	Love	N/A	N/A
2014/0069639	12/2013	Mackenzie et al.	N/A	N/A
2014/0151068	12/2013	Gilleylen et al.	N/A	N/A
2014/0190749	12/2013	Lorenson et al.	N/A	N/A
2014/0199196	12/2013	Twardowski	N/A	N/A
2014/0216761	12/2013	Trinh	N/A	N/A
2014/0246241	12/2013	Phillips	N/A	N/A
2014/0262216	12/2013	Perry	N/A	N/A
2015/0008043	12/2014	Clausen et al.	N/A	N/A
2015/0027708	12/2014	Honekamp	N/A	N/A
2015/0041217	12/2014	Gust	N/A	N/A
2015/0075867	12/2014	Eddison et al.	N/A	N/A
2015/0136403	12/2014	Cheng et al.	N/A	N/A

2015/0159437	12/2014	Crowley et al.	N/A	N/A
2015/0315902	12/2014	Beach et al.	N/A	N/A
2015/0354307	12/2014	Baudoin	N/A	N/A
2016/0032653	12/2015	James et al.	N/A	N/A
2016/0108677	12/2015	von Gynz-Rekowski et al.	N/A	N/A
2016/0281449	12/2015	Lorenson	N/A	N/A
2017/0159387	12/2016	Johnson	N/A	N/A
2017/0204693	12/2016	von Gynz-Rekowski et al.	N/A	N/A
2017/0254179	12/2016	Horwell	N/A	N/A
2018/0058145	12/2017	Tutt	N/A	N/A
2018/0156001	12/2017	Schwendemann et al.	N/A	N/A
2018/0163495	12/2017	Kinsella et al.	N/A	N/A
2018/0163496	12/2017	Oag et al.	N/A	N/A
2018/0171719	12/2017	Donald et al.	N/A	N/A
2018/0230750	12/2017	Lorenson	N/A	N/A
2018/0236580	12/2017	Kolbe	N/A	N/A
2018/0291733	12/2017	Ritchie et al.	N/A	N/A
2019/0010762	12/2018	Rossing et al.	N/A	N/A
2019/0257167	12/2018	Kinsella et al.	N/A	N/A
2020/0024924	12/2019	Clausen et al.	N/A	N/A
2020/0123856	12/2019	Sicilian et al.	N/A	N/A
2020/0141187	12/2019	Lorenson	N/A	N/A
2020/0240227	12/2019	Kinsella et al.	N/A	N/A
2022/0145714	12/2021	Kinsella et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2175296	12/1996	CA	N/A
2196857	12/1997	CA	N/A
2316796	12/1997	CA	N/A
2320280	12/1998	CA	N/A
2421211	12/2001	CA	N/A
2497458	12/2003	CA	N/A
2255065	12/2006	CA	N/A
2312341	12/2006	CA	N/A
2658162	12/2007	CA	N/A
2671171	12/2010	CA	N/A
2680895	12/2010	CA	N/A
2487380	12/2012	CA	N/A
2890072	12/2013	CA	N/A
2798807	12/2013	CA	N/A
2836182	12/2013	CA	N/A
2892971	12/2013	CA	N/A
2922999	12/2014	CA	N/A
2933482	12/2014	CA	N/A
2872736	12/2014	CA	N/A

2951397	12/2014	CA	N/A
2960699	12/2015	CA	N/A
2994473	12/2016	CA	N/A
105735929	12/2015	CN	N/A
0335543	12/1988	EP	N/A
3334891	12/2016	EP	N/A
713052	12/1953	GB	N/A
2059481	12/1980	GB	N/A
2398086	12/2003	GB	N/A
2529481	12/2015	GB	N/A
94/016189	12/1993	WO	N/A
2011058307	12/2010	WO	N/A
2012116393	12/2011	WO	N/A
2012/138383	12/2011	WO	N/A
2013106938	12/2012	WO	N/A
2014081417	12/2013	WO	N/A
2014089618	12/2013	WO	N/A
2015081432	12/2014	WO	N/A
2016024968	12/2015	WO	N/A
2016027103	12/2015	WO	N/A
2017027960	12/2016	WO	N/A
2018/006178	12/2017	WO	N/A

OTHER PUBLICATIONS

Drill on Target Directional Services Inc., “Drill on Target: Advanced Drilling Solutions”, published at least as early as Aug. 20, 2015 (5 pages). cited by applicant

Wedel, Ryan et al., “Mitigating Bit-Related Stick-Slip With a Torsional Impact Hammer”, 2011 AADE National Technical Conference and Exhibition held at the Hilton Houston North Hotel, Houston, Texas, Apr. 12-14, 2011, pp. 1-5. cited by applicant

Powell, Scott et al., “Fluid Hammer Increases PDC Performance through Axial and Torsional Energy at the Bit”, SPE 166433, SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, 30 Sep.-Oct. 2, 2013, pp. 1-7. cited by applicant

National Oilwell Varco, “6 3/4” FluidHammer—Operating Specifications”, published at least as early as Nov. 2015, p. 1. cited by applicant

NOV Wellbore Technologies, “Agitator(TM) Systems”, published at least as early as Feb. 2016, pp. 1-5. cited by applicant

International Search Report and Written Opinion dated Oct. 12, 2017 from PCT/CA2017/050828, pp. 1-11. cited by applicant

Office Action dated Apr. 23, 2015 from CA 2,872,736, pp. 1-3. cited by applicant

International Search Report and Written Opinion dated Feb. 23, 2015 from PCT/CA2014/051155, pp. 1-7. cited by applicant

International Preliminary Report on Patentability dated Jan. 17, 2019 from PCT/CA2017/050828, pp. 1-8. cited by applicant

Extended European Search Report dated Jan. 30, 2020 from EP 17823382.1, pp. 1-7. cited by applicant

International Search Report and Written Opinion dated Oct. 13, 2016 from PCT/CA2016/050950, pp. 1-7. cited by applicant

International Preliminary Report on Patentability dated Mar. 1, 2018 from PCT/CA2016/050950, pp. 1-5. cited by applicant

International Search Report and Written Opinion dated Oct. 28, 2016 from PCT/CA2016/051096,

pp. 1-8. cited by applicant
International Preliminary Report on Patentability dated Mar. 29, 2018 from PCT/CA2016/051096,
pp. 1-6. cited by applicant
Office Action dated Apr. 4, 2019 from U.S. Appl. No. 15/751,667, pp. 1-11. cited by applicant
Notice of Allowance dated Feb. 5, 2020 from U.S. Appl. No. 15/757,914, pp. 1-59. cited by applicant
Office Action dated Mar. 25, 2020 from U.S. Appl. No. 16/736,500, pp. 1-10. cited by applicant
International Preliminary Report on Patentability dated Jun. 16, 2016 from PCT/CA2014/051155,
pp. 1-5. cited by applicant
Office Action dated Oct. 4, 2016 from U.S. Appl. No. 14/771,418, pp. 1-19. cited by applicant
International Search Report and Written Opinion dated Sep. 9, 2016 from from
PCT/CA2016/050794, pp. 1-10. cited by applicant
International Preliminary Report on Patentability dated Mar. 1, 2018 from PCT/CA2016/050794,
pp. 1-8. cited by applicant
Extended European Search Report dated May 16, 2019 from EP 16836307.5, pp. 1-11. cited by applicant
Examiner's Report dated May 6, 2020 from EP 16836307.5, pp. 1-6. cited by applicant
Office Action dated May 10, 2018 from U.S. Appl. No. 15/892,866, pp. 1-23. cited by applicant
Office Action dated Jul. 15, 2019 from U.S. Appl. No. 16/382,610, pp. 1-39. cited by applicant
Office Action dated Aug. 17, 2018 from CA 2,798,807, pp. 1-5. cited by applicant
Office Action dated May 30, 2019 from CA 2,798,807, pp. 1-4. cited by applicant
Office Action dated Jun. 3, 2019 from CA 2,836,182, pp. 1-4. cited by applicant
Office Action dated Mar. 22, 2016 from U.S. Appl. No. 14/153,646, pp. 1-22. cited by applicant
Notice of Allowance dated Dec. 21, 2016 from U.S. Appl. No. 14/153,646, pp. 1-16. cited by applicant
Examination Report dated Mar. 18, 2021 from Australian Patent Application No. 2016308770, pp. 1-5. cited by applicant
Examination Report dated Jun. 28, 2021 from U.S. Appl. No. 16/849,055, pp. 1-23. cited by applicant
Examination Report dated Mar. 29, 2022 from AU2017292912, pp. 1-4. cited by applicant
Examination Report dated Sep. 27, 2022 from U.S. Appl. No. 17/586,087, pp. 1-20. cited by applicant
Examination Report dated Nov. 23, 2022 from CA2994474, pp. 1-3. cited by applicant
Notice of Allowance dated Mar. 7, 2023 from CA2994473, 1 pg. cited by applicant
Challenger Downhole Tools Inc., "Challenger Sidewinder", retrieved from
<https://web.archive.org/web/20150814052022/http://challengerdownhole.com/downhole-tool-products/challenger-sidewinder/>, Aug. 14, 2015, 1 pg. cited by applicant
Challenger Downhole Tools Inc., "Challenger Sidewinder", retrieved from
<https://web.archive.org/web/20160109112740/http://challengerdownhole.com:80/downhole-tool-products/challenger-sidewinder/>, Jan. 9, 2016, 2 pgs. cited by applicant
Cougar Drilling Solutions, "MT Mechanical Thruster Technical Specs", 1 pg (2020). Retrieved Feb. 27, 2025 from https://www.amasenergy.com/sites/default/files/attach/2020-08/cougar_ds_-_mt_mechanical_thruster_-_technical_specifications_1598863182_1598863223.pdf. cited by applicant
Energy Dais, "Mechanical Thruster(TM) by Cougar Drilling Solutions", 5 pgs. Retrieved Feb. 27, 202 from <https://www.energydais.com/cougar-drilling-solutions/mechanical-thruster-tm-7753/>.
Best available copy. cited by applicant
NOV Wellbore Technologies, "Agitator(TM) Systems", published at least as early as Feb. 2016, 5 pgs. cited by applicant
Rasheed, W. "Extending the Reach and Capability of Non Rotating BHAs by Reducing Axial

Friction.” Paper presented at the SPE/ICoTA Coiled Tubing Roundtable, Houston, Texas, Mar. 2001. doi: <https://doi.org/10.2118/68505-MS>. 9 pgs. cited by applicant

Weatherford International, “Renegade Friction-Reduction Tool”, 2 pgs (2013). Retrieved Feb. 27, 2025 from <https://www.weatherford.com/documents/technical-specification-sheet/products-and-services/completions/renegade-friction-reduction-tool/>. cited by applicant

Office Action dated Jul. 3, 2023 in Australian Patent Application No. 2022201161, pp. 1-5. cited by applicant

Office Action dated Feb. 7, 2019 in Canadian Patent Application No. 2994473, pp. 1-4. cited by applicant

Office Action dated Mar. 18, 2021 in Australian Patent Application No. 2016308770, pp. 1-5. cited by applicant

Office Action dated Oct. 12, 2018 from U.S. Appl. No. 15/892,866, pp. 1-11. cited by applicant

Office Action dated Oct. 3, 2023 in Canadian Patent Application No. 3029872, pp. 1-4. cited by applicant

Office Action dated Mar. 29, 2022 in Australian Patent Application No. 2017292912, pp. 1-4. cited by applicant

Office Action dated Mar. 23, 2023 in Australian Patent Application No. 2017292912, pp. 1-4. cited by applicant

Office Action dated Mar. 31, 2023 in U.S. Appl. No. 17/586,097, pp. 1-17. cited by applicant

Office Action dated Apr. 18, 2024 in U.S. Appl. No. 18/486,833, pp. 1-14. cited by applicant

Office Action dated Oct. 28, 2024 in U.S. Appl. No. 18/486,833, pp. 1-11. cited by applicant

Primary Examiner: Wallace; Kipp C

Attorney, Agent or Firm: Wilson Lue LLP (firm)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/197,896, filed Mar. 10, 2021, which is a continuation of U.S. patent application Ser. No. 16/241,029, filed Jan. 7, 2019, which is a continuation of International Application No. PCT/CA2017/050828, filed Jul. 7, 2017, which claims priority to U.S. Provisional Application No. 62/359,683, filed Jul. 7, 2016, the entireties of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to downhole drilling assemblies for use in horizontal and vertical drilling operations, and in particular valve control within a drilling string.

TECHNICAL BACKGROUND

(2) In oil and gas production and exploration, downhole drilling can be accomplished with a downhole drill powered by a mud motor. The drilling fluid used to drive the motor also assists the drilling process in other ways, for example by dislodging and removing drill cuttings, cooling the drill bit, and providing pressure to prevent formation fluids from entering the wellbore.

(3) Stalling and slip-stick issues can result in damage to drilling string components. It is believed that applying a vibrational or oscillating effect to the drill string components can improve performance of a downhole drill, and/or mitigate or reduce incidences of stalling and slip-stick.

(4) Further, when drilling deep bore holes in the earth, sections of the bore hole can cause drag or excess friction which may hinder proper weight transfer to the drill bit or causes erratic torque in the drill string. These effects may have the result of slowing down the rate of penetration, creating

bore hole deviation issues, or even damaging drill string components.

(5) Friction tools are often used to overcome these problems by vibrating a portion of the drill string to reduce friction or hole drag. Friction tools may form part of the downhole assembly of the drilling string, and can be driven by the flow of drilling fluid through the friction tool. Accordingly, the operation of a friction tool may be constrained by the flow rate of drilling fluid pumped through the string. Controlling the frequency of operation of the friction tool may therefore require varying or stopping the flow rate of the drilling fluid at the surface.

(6) It is not always desirable to run a friction tool during the entirety of a drilling operation. For instance, it may be unnecessary or undesirable to run the tool while the drill bit is at a shallow depth, or at other stages of the drilling operation where the added vibration of the friction tool is problematic or not required. During those stages, the drill string may be assembled without the friction tool. However, when a location in the bore hole is reached where the need for a friction tool is evident, it may then necessary to pull the downhole assembly to the surface to reassemble the drilling string to include the friction tool, then return the drilling string to the drill point. This process can consume several work hours.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In drawings which illustrate by way of example only embodiments of the present disclosure, in which like reference numerals describe similar items throughout the various figures,

(2) FIG. 1 depicts a portion of a drilling string including a flow-through pulsing assembly.

(3) FIG. 2 is a lateral cross-sectional view of the flow-through pulsing assembly of FIG. 1.

(4) FIG. 3 is a lateral cross-sectional view of a ball catch sub for use in the flow-through pulsing assembly.

(5) FIG. 4 is a lateral cross-sectional view of a flow-through shaft for use in the flow-through pulsing assembly.

(6) FIGS. 5A and 5B are side elevation and top views, respectively, of a stationary component of a rotating variable choke assembly for use in the flow-through pulsing assembly.

(7) FIGS. 6A, 6B, 6C, and 6D are a top view, a side elevation, a bottom view, and a lateral cross-sectional view, respectively, of a rotary component of the variable choke assembly.

(8) FIGS. 7A and 7B are a lateral cross-sectional view and an axial cross-sectional view, respectively, of the variable choke assembly when the stationary component and rotary component are in a first alignment.

(9) FIGS. 8A and 8B are a lateral cross-sectional view and an axial cross-sectional view, respectively, of the variable choke assembly when the stationary component and rotary component are in a second alignment.

(10) FIGS. 9 and 10 are cross-sectional views of the flow-through pulsing assembly with the ball catch sub in a first state and a second state of engagement, respectively.

(11) FIGS. 11A and 11B are side elevation and lateral cross-sectional views of an alternative variable choke assembly with a dart flow restrictor.

(12) FIGS. 12A and 12B are side elevation and lateral cross-sectional views of a ball catch assembly with a dart flow restrictor.

(13) FIG. 13 is a cross-sectional view of another configuration of the flow-through pulsing assembly.

(14) FIG. 14 is a lateral cross-sectional view of a portion of the flow-through pulsing assembly with a further variable choke assembly.

(15) FIGS. 15A and 15B are side and bottom views of the further variable choke assembly of FIG. 14;

(16) FIGS. **16A** and **16B** are lateral cross-sectional views of the further variable choke assembly taken along axes E and F, respectively, indicated in FIG. **15B**.

(17) FIGS. **17** and **18** are lateral cross-sectional views of the rotary and stationary components, respectively, of the further variable choke assembly.

(18) FIGS. **19A** and **19B** are a top view of the adaptor and bottom view of the stationary component, respectively, of the further variable choke assembly.

(19) FIGS. **20** and **21** are bottom and side views, respectively, of a portion of the flow-through pulsing assembly with the further variable choke assembly in a first alignment.

(20) FIGS. **22** and **23** are bottom and side views, respectively, of the portion of the flow-through pulsing assembly with the further variable choke assembly in a second alignment.

DETAILED DESCRIPTION OF THE INVENTION

(21) As is generally understood by those skilled in the art, in prior art downhole assemblies employing a power section (motor), drilling fluid passes from a bore or passage above the motor and into the motor to thereby activate the motor. This may be achieved by causing a rotor to rotate, and consequently drive any downhole tools linked to the rotor, such as a friction tool. Fluid passing through the motor enters the bore or passage downstream of the rotor. As can be seen in the particular example assembly **10** illustrated in FIGS. **1** and **2** and as discussed in further detail below, the drilling fluid passes from the motor section **200** to the drive section **300** and on to the valve section **400**; the rotor **210** is mechanically linked to the valve assembly in the valve section **400** to thereby drive a rotating component of the valve assembly.

(22) The rotation speed and horsepower of the motor is determined in part by the flow rate of drilling fluid through the motor. In a Moineau motor (“mud motor”), the particular lobe configuration of the motor and the drilling fluid type and properties will affect the motor output as well. In practice, once a drilling string is assembled and in place in the wellbore, the rotation speed and power of a motor such as a Moineau power section are changeable only by varying the flow rate of drilling fluid or else by retracting the drilling string from the bore hole, disassembling it, and reassembling it with a differently configured motor. However, it may not be desirable to vary the flow rate of the drilling fluid in this manner, and disassembling and reassembling a drilling string can consume several hours of labour.

(23) Accordingly, a flow-through assembly **10** with a selectively activatable motor and rotating variable choke assembly is provided for use in a downhole drilling string. The flow-through assembly **10** provides a system that can be inserted into the bore hole and then selectively activated or deactivated to control the flow of drilling fluid through the motor assembly, and from the motor to any tools or other features controlled or activated by the motor assembly. When the assembly **10** includes the rotating variable choke assembly, the variable choke assembly can be selectively activated or deactivated to provide a pulsing fluid flow for use in operating friction reduction tools or other types of tools. In these embodiments, activation of the motor can include starting the motor from a stopped or stalled state (i.e., no rotation of the rotor), to an “on” state in which the rotor rotates, or from a lower output state (i.e., a lower rate of rotation or lower torque output), to an increased or higher output state (i.e., a higher rate of rotation or higher torque output).

(24) The structure of the flow-through assembly **10** is generally illustrated in FIGS. **1** and **2**, which provide lateral views of one example of the assembly **10** with FIG. **1** being a view of the flow-through assembly **10** in relative position within a drilling string (indicated in phantom), and FIG. **2** being a lateral cross-sectional view of the flow-through assembly **10**. As can be seen in FIG. **1**, the exterior of the assembly **10** is defined by interconnected components **100**, **200**, **300**, and **400**, which may be provided as independent components to facilitate assembly and transport of the assembly **10** within a drilling string, and to further facilitate repair of the drilling string and/or the assembly **10** in the event of failure of an individual component of the assembly **10**. The components **100**, **200**, **300**, **400** can be connected using appropriate means, such as threaded connections. The assembly **10** or its individual components can be located in the drilling string above or below other

tools, not illustrated; for example, a shock sub or other tool providing oscillation or jarring effects may be disposed either below or above the assembly **10**.

(25) In the particular illustrated example, component **100** is a “ball catch” sub **100** comprising ball catch components used to catch and retain a ball dropped into the drilling string by an operator (as illustrated in FIG. **10**) above the rotor of the following motor section **200**. The third component **300** is an adaptor or drive section **300** used to transmit torque from the motor of the motor section **200** to the valve assembly comprised in the fourth component, the valve section **400**.

(26) Turning to FIG. **2**, the ball catch sub **100** includes a housing **105** encasing all or part of a ball catch head **110** and a ball catch seat **120**, both of which are retained within a ball catch retainer **130**. Each of these components is provided with a through bore **116**, **122**, **134**. A spring **138** or other biasing means is mounted on an interior shoulder **136** defined in a lower portion of the ball catch retainer **130**, within the bore **134**. A set of one or more bypass ports **140** may be provided in a wall of the ball catch retainer **130** above the interior shoulder **136**, to permit passage of fluid between the interior and exterior of the retainer **130**. An upper face **132** of the ball catch retainer **130** supports the ball catch head **110**. The ball catch head **110** includes a funnel-like opening **112** sized to receive and direct a ball towards the lower, substantially cylindrical portion of the ball catch head **110**. The wall of the funnel-like opening **112** is provided with the one or more bypass ports **114** that permit passage of fluid from the interior of the ball catch head **110** to its exterior. The funnel-like opening **112** is in fluid communication with the bore **116**. In the example of FIG. **3**, the exterior of the ball catch head **110** includes a circumferential flange component **118** that rests on the upper face **132** of the ball catch retainer **130**.

(27) The ball catch seat **120** is supported within the interior of the ball catch retainer **130**, below the ball catch head **110**. A lower face of the ball catch seat **120** rests on the spring **138**, and is able to reciprocate up and down within the ball catch retainer **130** as the degree of compression in the spring **138** changes under the force of drilling fluid flow when a ball **50**, as shown in FIG. **10**, is received on the ball catch seat **120**. The ball catch seat **120** is a substantially cylindrical component having a through bore **122** in fluid communication with the bore **134** of the ball catch retainer **130** and the bore **116** of the ball catch head **110**, and having a varying interior diameter or surface designed to catch a ball received from the ball catch head **110**. The ball catch seat **120** includes an interior shoulder or projection **124**. This interior shoulder defines a region of reduced interior bore diameter in the seat **120**, and is sized to retain an appropriately sized dropped ball in place and prevent its passage further downward.

(28) When the ball catch assembly is not engaged, fluid entering the ball catch assembly can pass through the ball catch head **110**, the bores **116**, **122**, and **134** and into other components of the assembly **10** below the ball catch assembly. Some fluid may pass through the bypass ports **114** and around the exterior of the ball catch assembly, but most fluid is expected to pass through the head **110** and bores. Thus, fluid entering the ball catch head **110** from above can pass down through the bore **116**, or through the bypass ports **114** and thus pass over the outside of the ball catch head **110** and the ball catch retainer **130**. When the ball catch assembly is engaged, a projectile such as the ball **50** blocks passage of fluid at the ball catch seat **120**; therefore, fluid entering the ball catch assembly will flow through the ports **114** and down around the exterior of the ball catch head **110** and retainer **130** in the space defined between these components and the housing **105**, and down to other components of the assembly **10** below the ball catch assembly that are in fluid communication with the exterior of the ball catch head **110** and retainer **130**.

(29) Other ball catch assemblies can be used in place of the ball catch sub **100** described above. Other examples of ball catch subs are described in International Applications No. PCT/CA2016/050950, “Selective Activation of Motor in a Downhole Assembly”, and PCT/CA2016/051096, “Selective Activation of Motor in a Downhole Assembly and Hanger Assembly”, the entireties of which are incorporated herein by reference. Furthermore, implementations of the flow-through assembly **10** may exclude a ball catch sub positioned above

the valve section **400**.

(30) In the example assembly **10** shown in FIG. 2, rotor **210** is provided with a bore **212** extending through the length of the rotor **210**, and the bore **134** is in fluid communication with the bore **212**. In the illustrated example, the rotor **210** and ball catch assembly are directly joined by a threaded connection, but they may be connected by an intermediate unit, such as the shaft **310** described below. The illustrated shaft **310** may be referred to as a flow-through shaft, flow-through drive shaft, or flex shaft. For convenience, the shaft **310** is generally referred to as a drive shaft **310** below.

(31) Returning to FIG. 2, the motor section **200** includes a cooperating stator **205** and rotor **210**. In the example assembly **10** depicted here, the motor is a Moineau motor, with a multi-lobe rotor **210** rotating in a multi-lobe stator. The rotor **210** in this example, as mentioned above, includes a through bore or passage **212** providing for fluid communication from the bore **134** of the ball catch retainer **130**.

(32) The drive section **300** comprises a housing **305** enclosing at least a substantial part of a flow-through drive shaft **310**, thus defining an annular space between the interior diameter of the housing **305** and the outer diameter of the drive shaft **310**. The drive shaft **310**, which is illustrated in further detail in FIG. 4, comprises a substantially elongated main body **312** with a through bore **314** to permit passage of fluid therethrough. An upper end of the drive shaft **310** is connected to the lower end of the rotor **210**, while the lower end of the drive shaft **310** is connected to an upper end of the valve assembly in the valve section **400**, and specifically an upper end of the rotary valve component **410**. As the bore **314** of the flow-through drive shaft **310** provides for fluid communication between the rotor bore **212** and the variable choke assembly below, suitable joints or connections are provided between the drive shaft **310** and the rotor **210** and the drive shaft **310** and the rotary component **410** to permit fluid communication therethrough. In the particular example illustrated in the accompanying figures, the drive shaft **310** is joined to both the rotor **210** and the rotary valve component **410** by threaded connections **316** to minimize obstruction of any fluid passing through the bore **314**. The portions of the drive shaft **310** between the main body **312** and the threaded connections **316** may be enlarged (e.g., with greater wall thickness than the elongated main body **312**) to increase the strength of the drive shaft **310** at those points, while still providing the annular space between the exterior of the drive shaft **310** and the interior of the housing **305**. For instance, in one non-limiting example, the outer diameter of the drive shaft **310** at the enlarged portions near the threaded connections can be about 2.25 inches, tapering to about 1.825 inches for the rest of the main body **312**, while maintaining an interior bore diameter of about 1.5 inches throughout.

(33) Returning again to FIG. 2, the valve section **400** includes a housing **405** enclosing the aforementioned rotary component **410** connected to the flow-through drive shaft **310**. The rotary component **410** rotates under influence of the rotor **210** within a radial bearing **440** and on a rotary bearing **450** situated in the housing **405**. Flow ports **424** provided in the body of the rotary component **410** enter into and out of engagement with a corresponding stationary component **430**, also housed in the housing **405**.

(34) The stationary and rotary components **430**, **410** are illustrated in further detail in FIGS. 5A to 6D. Turning first to FIGS. 5A and 5B, the stationary component **430** comprises a substantially annular component sized to fit within the valve section housing **405**, and to receive the rotary component **410** within the stationary bore **434**. The interior face **436** of the stationary component **430** provides the bore **434** with a substantially cylindrical configuration, with one or more channels **438** creating regions of increased bore diameter. The diameter of the bore **434** is sized to fit the rotary component **410** and to permit fluid access to the flow ports **424** of the rotary component **410** when the flow ports **424** are at least partially coincident with a corresponding channel **438**, and to substantially block fluid access when the channels **438** are not coincident with the ports **424**, as shown in further detail with reference to FIGS. 7A to 8B.

(35) FIG. 6A illustrates a side elevational view of the rotary component **410**, while FIG. 6B provides a view of the cross-section of the view of FIG. 6A taken along plane A-A, and FIGS. 6C and 6D illustrate top and bottom view of the rotary component **410**, respectively. The rotary component **410** in this particular example is substantially cylindrical or bullet-shaped, with a slightly tapered upper portion. The body of the rotary valve component **410** includes a bore **416** extending from the bottom to the top of the component **410**, thus providing for fluid flow straight through the body. The rotary component **410** also includes at least one bypass port **422** and at least one flow port **424**, which provide for fluid communication between an exterior of the rotary valve component **410** and the bore **416**. As can be best seen in FIGS. 6A and 6B, the outlets of the bypass ports **422** on the exterior surface of the component **410** are disposed within recessed facets **420** of the valve component's exterior. These facets originate at a midsection of the component **410** and extend towards the top of the component **410** at an incline, such that they are angled towards the centre of the body (i.e., towards the bore **416**) at towards the top of the component **410**. This provides a slightly tapered profile to the generally cylindrical shape of the component **410**, such that the circumference or perimeter at the top of the component **410** is smaller than at a point around the midsection of the component **410**.

(36) The flow ports **424** are provided at or around the midsection of the rotary valve component **410**, and are generally laterally aligned with the bypass ports **422**; as can be seen in the illustrated examples, the flow ports **424** are located directly below the bypass ports **422**. As may be better appreciated with reference to FIG. 8A, this permits drilling fluid flowing downwards in the annular space between the drive shaft **310** and the interior of the housing **305**, **405** to enter into the bypass ports **422**, as well as the flow ports **424** of the rotary component **410**, provided access to the flow ports **424** are not blocked by the stationary component **430** as discussed below.

(37) Fluid access to the bypass ports **422** and flow ports **424** from above the rotary component **410** can be enhanced by further angling or tapering of the upper portion of the component **422**; for example, the remaining upper exterior surfaces **418** of the component **410** are likewise angled towards the top of the component **410**, as can be seen in FIGS. 6A and 6B.

(38) FIGS. 7A and 7B illustrate the variable choke assembly in a “choked” position, while FIGS. 8A and 8B show the variable choke assembly in an “open” position. The rotary component **410** can enter into and out of these positions as it rotates inside the stationary ring component **310** while driven by the rotor **210**; when the rotor **210** is not rotating, the rotary component **410** may be positioned in the “open” position, the “choked” position, or an intermediate position. If the rotor is in a lower output state (lower rate of rotation or output torque), the rotary component **410** will move between the “open” and “choked” positions. As can be seen in FIG. 7A, the rotary component **410** rests and rotates on the rotary bearing **450** disposed within the valve section housing **400**. The rotary bearing **450** is substantially annular and thus permits passage of drilling fluid from the bore **416** of the rotary component **410** to the components of the drilling string below the valve section **400**. The stationary component **430** surrounds the rotary component **410** around the midsection of this latter component at about a level of the flow ports **424**; the bypass ports **422** are positioned above the stationary component **430**.

(39) In the “choked” or “restricted” position, the outlets of the flow ports **424** are substantially blocked because the interior face **436** of the stationary component **430** contacts the exterior of the rotary component **410** above the flow ports **424**, thereby cutting off fluid access to the flow ports **424**. However, even in the “choked” state, the bypass ports **422** will still remain unblocked since the outlets of those ports **422** are disposed on a recessed upper portion of the rotary component **410**, as discussed above. In addition, regardless whether the variable choke assembly is in the “choked” or “open” state, the bore **416** still permits passage of drilling fluid, drilling string instruments, and blocking projectiles to the downhole portions of the drilling string (assuming that the ball catch assembly is not engaged and blocking through passage), even when the rotary component **410** is rotating.

(40) In the “open” position, as shown in FIGS. 8A and 8B, the flow ports **424** are substantially aligned with the channels **438** in the stationary component **410**; thus, fluid can enter into the channels **438** and thence into the flow ports **424** and the bore **416**. In a partially “open” position, the flow ports **424** are only partially aligned with the channels **438**, so less fluid can enter the channels **438** and the flow ports **424**. The bypass ports **422** remain open because the outlets of the ports **422** are disposed on a recessed portion of the rotary component **410** above the stationary component **430**. The flow rate through the flow ports **424** can be adjusted by altering the interior dimensions and distribution of the flow ports **424** around the rotary component **410**, and/or by altering the dimensions of the recesses **438** in the stationary component **430**. For example, the interior dimensions of the flow ports **424** can be reduced with an optional lining, such as a carbide insert (not shown).

(41) The operation of the flow-through assembly **10** can be understood by referring to FIGS. 9 and 10, which illustrate the effect on drilling fluid flow when the rotating variable choke assembly is activated. In FIG. 9, the ball catch assembly is not in an engaged state. No projectile **50** is in place in the ball catch seat **120**; consequently, drilling fluid entering the ball catch assembly from above can flow into the bore **134** of the ball catch retainer **130** and into the bore **212** of the rotor **210**, as indicated by arrows in FIG. 9. The fluid exits the bore **212** and passes through the bore **314** of the drive shaft **310**, and the bore **416** of the rotary component **410**. Some drilling fluid may still flow around the exterior of the ball catch retainer **130** and enter the motor. Since most fluid enters the bore **212**, the rotor **210** will be either stalled or in a low output state.

(42) The fluid then passes into the bore **416** of the rotary component **410**. Most drilling fluid entering the ball catch assembly will pass through the centre bore **212** of the rotor, then bores **314** and **416**. However, if any fluid happens to reach the exterior of the rotary component **410**, it may enter one of the bypass ports **422** and enter the bore **416** in that way; and if the rotary component **410** is in an “open” or partially-“open” position, some fluid may even enter the bore **416** via the flow ports **424** to the extent they are not blocked off. Thus, when the ball catch assembly is in the non-engaged state, the substantial part of the drilling fluid flows through the communicating bores of the various components with minimal variation in fluid pressure.

(43) On the other hand, when the ball catch assembly is in the engaged state as in FIG. 10, a ball **50** or other blocking projectile is seated in the ball catch seat **120**. This causes drilling fluid to be substantially blocked from passing through the bore **134**. As indicated by the arrows in FIG. 13, drilling fluid is therefore directed from the ball catch head **110**, through the ports **114** in the funnel **112**, and down the exterior of the ball catch retainer **130** toward the cavities of the motor defined by the rotor **210** and stator **205**. This provides sufficient flow to activate the motor, causing rotation of the rotor **210**, or to significantly increase the output of the motor, thereby driving the rotary component **410** of the variable choke assembly (at a higher rate). Minimal fluid will pass through the rotor bore **212** and drive shaft bore **314**. The drilling fluid exiting the motor passes around the exterior of the drive shaft **310** and the exterior of the rotary component **410**, which is rotating. Some fluid will enter the bypass ports **422** of the rotary component **410**, while other fluid will intermittently enter the flow ports **424** as rotary component **410** rotates and the flow ports **424** move into and out of alignment with the channels **438** in the stationary ring component **430**, as indicated by the phantom arrows in FIG. 10.

(44) The varying rate of fluid consequently entering the bore **416** will produce variations in the fluid pressure above the rotary component **410**. The fluid pressure will vary between a minimum and maximum value, as the rotary valve component **410** rotates from the “choked” to “open” position. The resultant pressure variations can be used to operate an oscillation, friction, or impulse tool in the drilling string. It will be appreciated that even while pressure variations are being generated by the variable choke assembly, the assembly **10** still permits a significant amount of fluid to flow downstream to other drilling string components, such as the bottom hole assembly. This is because the rotary component of the variable choke assembly includes the bypass ports **422**,

permitting drilling fluid to bypass flow ports **424** even when the flow ports **424** are closed.

(45) Where the assembly **10** as depicted in FIG. **1** is included in a drilling string, an oscillation or impulse tool may be mounted either uphole, above the assembly **10**, or downhole, below the assembly **10**. The variations in fluid pressure caused by the operation of the rotary variable choke assembly may be transmitted a distance uphole, beyond the ball catch assembly, for example. Furthermore, it will be appreciated by those skilled in the art that in some drilling string arrangements, the various components of the assembly **10** can effectively be arranged in reverse order, with the valve section **400** uphole of the ball catch component **100** or a variant of the ball catch component **100**. FIG. **13** illustrates an example arrangement of an assembly **10** in which the rotary component **410** and stationary component **430** of the rotating variable choke assembly are retained in an inverted position at a top end of the assembly **10**. The rotary component **410** is connected to a ball catch assembly; FIG. **13** illustrates a simple version having a ball catch seat **120** without a funnel-like ball catch head, since a projectile would first pass through the bore **416** of the rotary component **410**, so the rotary component functions as the ball catch head. The ball catch assembly, in turn, is in fluid communication with the bore **212** of the rotor **210**, which is positioned below the rotary component **410** and the ball catch assembly. In this example, the ball catch assembly and the rotor **212** are connected by a flow-through drive shaft **310**, which provides for fluid communication through its bore **314** and also transmits torque generated by the rotor **210** to the ball catch assembly and rotary component **410**.

(46) When the ball catch assembly is not engaged, no projectile **50** is in place on the ball catch seat **120**, and drilling fluid entering the rotary component **410** passes through the rotary component bore **416**, the ball catch assembly, the drive shaft bore **314**, the rotor bore **212** in a manner similar to that described above. Minimal pressure variation is produced by the assembly **10**. When the ball catch assembly is engaged, the projectile blocks passage of drilling fluid down the central bores **314** and **212**. Drilling fluid enters the bore **416** from above, but the blockage of the bores **314** and **212** causes fluid to flow out through the bypass ports **422**, which remain unblocked as described above, and through the ports **424** provided exit from the ports **424** is not blocked by the stationary component **430**. This results in drilling fluid flow downwards around the exterior of the drive shaft **310**, and into the motor. This activates the motor, generating torque, which is transmitted from the rotor **210** to the ball catch assembly and rotary component **410** by the drive shaft **310**. As the rotary component **410** rotates, it will move between the “choked” and “open” positions described above, thereby varying the fluid pressure above the rotary component **410**. Again, the pressure variations generated by the assembly **10** can be used to operate an oscillation, friction, or impulse tool.

(47) In some implementations, the ball **50** can be manufactured of a durable, shatter-resistant material, such as stainless steel. In that case, once in place, the ball **50** is removable by retracting the assembly **10** to the surface, and disassembling a sufficient portion of the assembly **10** to retrieve the ball **50**. If the ball **50** has a sufficiently magnetic composition, then the ball may be retrieved by passing a rod or probe with a magnet affixed thereto to attract and withdraw the ball **50** from the assembly.

(48) In other implementations, the ball **50** can be manufactured of a breakable material, such as Teflon®. When such a ball **50** is in place as in FIG. **10** and the motor is active, the motor can be substantially stopped or slowed down by dropping a fracture implement (not shown), such as a smaller steel ball, to shatter to the ball **50** without retracting the assembly **10** to the surface. If the fracture implement has a smaller diameter than the various bores of the components in the assembly **10**, it may pass through the assembly **10** without substantially blocking fluid flow therethrough. Thus, it could be possible to selectively engage and disengage the ball catch sub **100**, thereby activating or deactivating the motor section **200** and the valve section **400** as desired to selectively provide a pulsing fluid flow through the drilling string.

(49) It will be appreciated by those skilled in the art that modifications can be made to the ball catch component **100**. For example, as shown in FIGS. **11A** and **11B**, the operation of the ball catch

component **100** can be effectively integrated into the valve section **400**. FIG. **11A** shows a side elevational view of the modified rotary component **410'** with a dart plug **500** seated therein. FIG. **11B** shows a cross-sectional view of this modified component **410'** and plug **500** taken along axis D-D. The modified component **410'** includes an interior seat **411** defined by the interior diameter of the component **410'**, which is sized and shaped to receive a corresponding seating portion **504** of the plug **500**. The plug **500** includes a leading end **502** and an opposing head end **506**. The leading end **502** in this example is tapered to a tip; the seating portion **504**, which is located between the ends **502** and **506**, is an exterior diameter tapering in size towards the leading end **502**. The overall shape of the plug **500**, particularly as defined by tapered profile of the leading end **502** and the seating portion **504**, assists in seating the plug **500** in the modified valve component **410'** when it is dropped into the drilling string. Seals may be provided on the exterior of the plug **500** to engage the interior wall of the modified valve component **410'**, so as to prevent drilling fluid flow around the plug **500**. Optionally, the head end **506** of a plug **500** can be provided with a hook or hole that is capable of being engaged by a wireline tool so that the plug **500** can be retracted through the drilling string without requiring disassembly.

(50) In the foregoing example, plug **500** is received in what was previously described as the upper portion of the rotary component **410**, above. Thus, in this modified example, end of the modified component **410'** is connected to a rotor at the opposing end. When assembled in the drilling string, the valve section containing the modified valve component **410'** would be located uphole from the motor section **200**, rather than downhole as illustrated in the earlier example. In this example, the ball catch component **100** is not required; the modified valve component **410'** operates to selectively activate or deactivate an oscillation or impulse tool in the string.

(51) Another variant in the ball catch component **100** is illustrated in FIGS. **12A** and **12B**. In this example, rather than provide separate ball catch head **110**, ball catch seat **120**, and ball catch retainer **130** components, a single integrated ball catch unit **510** is provided, similar to the ball catch described in U.S. Provisional Application No. 62/220,859, which is incorporated herein by reference. The dart is received in the ball catch unit **510** and sits against an interior seat **511**, similar to the interior seat **411** depicted in FIGS. **11A** and **11B**.

(52) FIGS. **14** to **23** illustrate a further embodiment of the variable choke assembly **600** that can be used with the flow-through pulsing assembly described above, or in other assemblies requiring a pulsing or variable fluid flow driven by a rotor. It will be appreciated by those skilled in the art that despite the inclusion of seals in a downhole assembly, some leakage may occur. Where two components rotate against each other, as in rotary valves or rotary choke assemblies such as the variable choke assembly described above, some leakage can occur during rotation due to slight transverse motion of one component, which may be due to the eccentric orbit of the rotor driving the rotational motion. Leakage of drilling fluid can result in an undesired drop in fluid pressure downstream of the leakage points. These drops in fluid pressure may require an increase in fluid pressure at the surface to compensate, but this in turn may accelerate wear on components upstream from the leakage points. Thus, in the embodiment of FIGS. **14** to **23**, the rotary and stationary components of the variable choke assembly are provided with complementary tapered faces that reduce leakage due to transverse motion.

(53) FIG. **14** depicts the relevant components of the variable choke assembly below the drive shaft **310**. A stationary component **650** of the variable choke assembly with a through bore **656** receives a corresponding rotary component **630** with a corresponding through bore **636**. As can be seen from the following figures, the rotary component **630** and stationary component **650** engage each other with complementary tapered surfaces. In the embodiment illustrated in FIGS. **14** to **23**, the rotary component **630** is mounted to the end of the drive shaft **310** by means of an adaptor shaft component **610**, which is also provided with a through bore **616**. At one end, the bore of the adaptor shaft component **610** can be threaded for connecting to the drive shaft **310**; the other end can be threadedly connected to the rotary component **630**. The rotary component **630** rotates on the

stationary component **650** within a radial bearing **620** mounted within the housing of the downhole assembly, as can be seen in FIG. **15A**.

(54) FIGS. **15A** to **16B** illustrate the assembled adaptor shaft component **610**, rotary and stationary components **630**, **650**, and radial bearing **620**. These components can be manufactured from a carbide; the adaptor shaft component **610** may be manufactured from stainless steel. In addition to their corresponding bores **616**, **636**, **656**, each of the adaptor shaft component **610**, rotary component **630**, and stationary component **650** are provided with ports that can enter into and out of alignment with each other as the rotary component **630** rotates against the stationary component **650**.

(55) The stationary component **650** is provided with one or more ports **652** passing through the body of the component **650**, around the through bore **656**. The ports are aligned to be substantially, but not necessarily, parallel to the through bore **656**. The cross-sectional shape and area of each port **652** may be the same, or different, depending on the desired pulsing effect of the variable choke assembly **600**. Similarly, they need not be spaced in regular intervals around the bore **656**. In the illustrated embodiment, each port **652** has a rounded arcuate cross-sectional opening, as discussed below. The rotary component **630** is provided with one or more ports **632** in its body, spaced around the through bore **636**. Again, the ports in the rotary component **630** need not be identically shaped or regularly spaced around the through bore **636**, depending on the desired pulsing effect; but in this example, the ports are identically shaped and arranged at regular intervals around the bore **636**. The ports **632** have a cross-sectional shape similar to, but shorter in length than, the ports **652** in the stationary component **650**. As can be seen in FIG. **16A**, the adaptor shaft component **610** is provided with corresponding ports **612** which align with the ports **632** of the rotary component **630** when these two components are joined together. In some implementations, the rotary component **630** can include an adaptor for mounting to the end of a drive shaft **310** or other component, thereby avoiding the need for a separate adaptor shaft component **610**.

(56) In the embodiment illustrated in the figures, the adaptor shaft and rotary components **610**, **630** are also provided with at least one bypass port **614**, **634** respectively. These ports **614**, **634** also align with each other when the adaptor shaft component **610** is mounted to the rotary component **630**. A carbide insert **615** is inserted in the bypass port **614** to reduce its circumference to control flow through the bypass port **634**. In the illustrated embodiment, four bypass ports **614**, **634** alternate with the ports **612**, **632**. In the illustrated configuration, when the ports **652** and **632** are in complete alignment, as illustrated by the bottom view of FIG. **15B** and FIG. **16A**, the bypass ports **614**, **634** are blocked by the solid body of the stationary component **650**, as shown in FIG. **16B**.

(57) FIGS. **17** and **18** show the rotary and stationary components **630**, **650** in isolation. In these views, the tapered bottom surface **633** of the rotary component **630** can be clearly seen. The bottom surface **633** is effectively inclined upward from the centre of the component **630** (i.e., the portion of the component comprising the through bore **636**) towards the outer edge of the component **630**. In this example, the incline is a 15 degree angle. The stationary component **650** is provided with an upper surface **653** with a complementary inclination downward from the edge of the component **650** towards the centre. Thus, when assembled, the rotary component **630** sits in the stationary component **650**. As the rotary component **630** rotates in the stationary component **650**, the ports **632** and **652** move into and out of alignment with each other; similarly, the bypass ports **624** move out of and into alignment with the ports **652**. As the rotary component **630** rotates, the inclined or tapered shape of the interface between the two components **630**, **650** reduces transverse or sideways travel, since the upper surface **653** of the stationary component **650** interferes with transverse movement of the rotary component **630**.

(58) FIG. **19A** illustrates the arrangement and shape of the ports **612** and/or **632**, and the bypass ports **624** and/or **624** of the adaptor shaft and rotary components **610**, **630**, while FIG. **19B** illustrates the arrangement and shape of the ports **652** in the stationary component **650**. In the illustrated example, the ports **612**, **632**, **652** have a cross-section that may be described as a slightly

arcuate ring section with rounded corners, or a kidney shape with flattened leading edges (see for example **615** and **655**). The bypass ports **614** may have a similar shape, but in this embodiment, have a circular cross-section. The bypass ports **614** and ports **612**, **632** in the rotating components have a smaller cross-sectional area than the stationary component ports **652**. The cross-sections of the ports **612**, **632**, in particular, are shorter in length than the cross-sections of the ports **652**, such that the entire cross-section of the ports **612**, **632** will intersect with the cross-section of the stationary component ports **652** for a period of time as the rotary component **630** (and adaptor shaft component **610**) rotates in the stationary component **650**. This provides additional time for the rotary/adaptor shaft components **630/610** to dump the fluid within their ports **632/612** before the ports move out of alignment. The flat leading edges **615**, **655** of the ports maximize the cross-sectional area available to permit fluid flow as the ports move into and out of alignment. As the person skilled in the art would appreciate, if the ports of the rotary and stationary components had a circular cross-section, as they move into and out of alignment the intersection of the ports would define a small biconvex lens shape, increasing to an circular shape, then immediately reducing to a small biconvex lens shape again. In other words, minimal time would be spent with the ports in maximal alignment. By providing larger ports **652** in the stationary component **650**, the ports **632** of the rotary component **630** will remain in maximal alignment with the ports **652** for longer than if the ports **632**, **652** were the same size. In addition, by squaring off the leading edges (and optionally trailing edges, as illustrated in the drawings), the ports **632**, **652** provide for more throughput as they move into and out of alignment.

(59) FIGS. **20** to **23** illustrate this variable choke assembly in first and second alignments within a drilling string. As illustrated in FIG. **20**, in a first alignment, or “choked” or “restricted” state, fluid flow through the entire assembly is restricted by the bypass ports **614/624**, which intersect the ports **652** of the stationary component **650**. The size and position of the bypass ports and other ports in the rotary/adaptor shaft components **630/610** can be selected so that at least one port of the rotary/adaptor shaft components is at least partially aligned with a port **652** at any time; although in other embodiments, all ports may be completely blocked at some point during rotation. When the variable choke assembly is in this “choked” state, fluid flow may be restricted as shown in FIG. **21**. If fluid is flowing down the bore **314** of the drive shaft **310**, it will pass through the corresponding bores of the variable choke assembly. Fluid passing on the outside of the drive shaft **310** (i.e., fluid that did not bypass the motor, as shown in FIGS. **9** and **10**) will enter the bypass ports **614/624** and exit through the stationary ports **652** when they are aligned.

(60) FIG. **22** illustrates a second alignment, or “open” state, when the ports are maximally aligned, enabling as much drilling fluid as possible to be dumped through the ports **652**. As shown in FIG. **23**, fluid flow through the variable choke assembly will be at its greatest when the ports are all aligned. Thus, it will be appreciated that as the rotary/adaptor shaft components rotate with respect to the stationary component, fluid flow will vary between a minimum and maximum value, providing a resultant variation in fluid flow and pressure. The shapes of the ports increase the pressure differential between the “choked” state (when fluid is maximally blocked) and the point at which the ports **632**, **652** enter into alignment, because they are shaped to provide as much instantaneous fluid flow as possible, and thus a greater pressure variation without requiring increased fluid pressure at the surface, thus potentially reducing wear on components in the drilling string, particularly when combined with the tapered configurations of the stationary and rotary components.

(61) Those skilled in the art will appreciate that the foregoing examples not only provide for selective activation of tools in the drilling string by permitting the operator to selectively activate, and optionally deactivate, the valve section **400** using the ball catch component **100**, but also provides a pathway for other tools and components to pass through the entire assembly **10** to downhole locations. The ball catch component **100**, motor section **200**, drive section **300**, and valve section **400** all provide a substantially continuous pathway, which can be adequately sized to permit

wireline gear to pass through the entire assembly **10** while it is still downhole. In addition, the pathway can permit the passage of other balls or similar projectiles through the assembly **10** and down to other tools located below the assembly **10**, such as other ball catch components, friction reduction tools, PBL subs, lost circulation subs, jars, reamers and the like.

(62) Furthermore, the examples provided above provide for selective activation and deactivation by creating a pathway for the bypass of drilling fluid through the assembly with components that present less of an obstacle to fluid flow in the drilling string as compared to the prior art. As those skilled in the art appreciate, fluid pressure and flow in drilling is critical to successful removal of cuttings from the wellbore, and to successful operation of the drill bit and other pressure-dependent tools in the string. While a number of factors impact the flow rate within a well, such as drilling fluid properties, system and formation pressure limits, the inclusion of different components in the drilling string restricting the effective cross-sectional area of the pathway available for fluid flow can impede the drilling operation by causing pressure drops in the system. Prior art solutions providing for fluid bypass can include several “layers” of cooperating components that effectively reduce the cross-section available for drilling fluid flow. The examples described above, on the other hand, provide a more optimal use of the cross-sectional space in the drilling string. Moreover, the examples above can function satisfactorily without altering the flow rate of drilling fluid into the assembly **10**.

(63) Throughout the specification, terms such as “may” and “can” are used interchangeably and use of any particular term should not be construed as limiting the scope or requiring experimentation to implement the claimed subject matter or embodiments described herein. Various embodiments of the present invention or inventions having been thus described in detail by way of example, it will be apparent to those skilled in the art that variations and modifications may be made without departing from the invention(s). The inventions contemplated herein are not intended to be limited to the specific examples set out in this description. For example, where appropriate, specific components may be arranged in a different order than set out in these examples, or even omitted or substituted. As another example, the number, sizes, and profiles of the ports **424**, **422** in the rotary valve component **410** and the corresponding recesses **438** in the stationary valve component **430** can be varied as appropriate to accomplish a desired frequency or pulsation effect, or to accommodate particular equipment or drilling fluid. The inventions include all such variations and modifications as fall within the scope of the appended claims.

Claims

1. An assembly for use in a tubular string, the assembly comprising: a motor comprising a rotor and a stator; a rotary component and a stationary component for generating fluid pressure pulses, the rotary component and stationary component each comprising at least one respective port, the rotary component being drivable by the rotor; the rotary component, stationary component, and rotor each comprising a bore defining a substantially continuous passage permitting a substantially axial first route for fluid flow through the assembly, the at least one port of the rotary component substantially aligned with the bore of the rotary component and the at least one port of the stationary component substantially aligned with the bore of the stationary component; the motor and the ports of the rotary component and stationary component providing a passage permitting a substantially axial second route for fluid flow through the assembly, wherein fluid passing through the assembly along the second route does not flow through the bore of the rotary component or the bore of the stationary component, wherein the passage through the motor is between the rotor and the stator; wherein when fluid flows in the assembly through both the first and second routes, the rotary component is driven at a first rotational speed; wherein when at least some of the fluid is diverted from the first route to the second route by a blockage disposed in the first route, the rotary component is driven at a second rotational speed.

2. The assembly of claim 1, wherein when at least some of the fluid is diverted from the first route to the second route by the blockage, the fluid flows through both the first and second routes.
3. The assembly of claim 1, wherein rotation of the rotary component with respect to the stationary component produces variations in fluid pressure depending on a rotational speed of the rotary component.
4. The assembly of claim 1, wherein the blockage comprises a blocking implement.
5. The assembly of claim 1, wherein the blocking implement comprises a plug, ball, or dart.
6. The assembly of claim 5, wherein the blocking implement comprises a dart.
7. The assembly of claim 1, wherein each of the rotary component and the stationary component comprise a plurality of respective ports.
8. The assembly of claim 7, wherein the plurality of ports of the rotary component are spaced around the bore of the rotary component, and the plurality of ports of the stationary component are spaced around the bore of the stationary component.
9. A method of operating a downhole assembly, wherein the downhole assembly comprises: a motor comprising a rotor and a stator; a rotary component and a stationary component for generating fluid pressure pulses, the rotary component and stationary component each comprising at least one respective port, the rotary component being drivable by the rotor; the rotary component, stationary component, and rotor each comprising a bore defining a substantially continuous passage permitting a substantially axial first route for fluid flow through the assembly, the at least one port of the rotary component substantially aligned with the bore of the rotary component and the at least one port of the stationary component substantially aligned with the bore of the stationary component; the motor and the ports of the rotary component and stationary component providing a passage permitting a substantially axial second route for fluid flow through the assembly, wherein fluid passing through the assembly along the second route does not flow through the bore of the rotary component or the bore of the stationary component, wherein the passage through the motor is between the rotor and the stator, the method comprising: causing fluid to flow into the downhole assembly through the first and second routes to thereby drive the rotary component at a first rotational speed; diverting at least some fluid flowing through the first route to the second route using a blocking implement to thereby drive the rotary component at a second rotational speed.
10. The method of claim 9, wherein diverting at least some fluid comprises blocking the continuous passage by a blocking implement disposed in the first route.
11. The method of claim 9, wherein fluid flows through both the first route and the second route while the blocking implement is disposed in the first route.
12. The method of claim 9, wherein rotation of the rotary component with respect to the stationary component produces variations in fluid pressure depending on a rotational speed of the rotary component.
13. The method of claim 9, wherein a catch is provided at a first end of the substantially continuous passage for receiving the blocking implement.
14. The method of claim 13, wherein the catch comprises at least one bypass port between the first end and the rotor.
15. The method of claim 9, wherein the blocking implement comprises a plug, ball, or dart.
16. The method of claim 15, wherein the blocking implement comprises a dart.
17. An assembly for use in a downhole string, the assembly comprising: a motor comprising a rotor and a stator; a rotary component and a stationary component for generating fluid pressure pulses, the rotary component and stationary component each comprising at least one respective port; the rotary component, stationary component, and rotor each comprising a bore defining a substantially continuous passage permitting a substantially axial first route for fluid flow through the assembly; the motor and the ports of the rotary component and stationary component providing a passage permitting a second route for fluid flow through the assembly, wherein fluid passing through the

assembly along the second route does not flow through the bore of the rotary component or the bore of the stationary component, wherein the passage through the motor is between the rotor and the stator, the rotary component being drivable by the rotor such that the at least one port of the rotary component enters into and out of alignment with the at least one port of the stationary component to thereby cause pulsing in fluid flowing in the second route in a direction substantially parallel to the axial first route; wherein when fluid flows in the assembly through both the first and second routes, the rotary component is driven at a first rotational speed; wherein when at least some of the fluid is diverted from the first route to the second route by a blockage disposed in the first route, the rotary component is driven at a second rotational speed.
